

Ethical and Empathetic Requirements for Trustworthy ChatBots

Sophia Jit
University of Toronto
Toronto, Canada
sophia.jit@mail.utoronto.ca

Nikita Dvornik
Palona AI
Montreal, Canada
nikita@palona.ai

Abstract—As large language model (LLM)-based chatbots become prevalent across domains such as healthcare, education, and customer service, ensuring their behavior aligns with normative requirements (NR) becomes essential for fostering trust in these systems. Yet, eliciting these requirements for such LLM-based chatbots remains challenging, due to their open-ended capabilities. In this paper, we propose NR that describe how chatbot responses should behave to be compliant with the norms, across varying inputs, contexts, and user profiles in three application domains: therapy, education, and customer service.

I. INTRODUCTION

With the growing integration of chatbots in domains such as healthcare, education, and customer service, we argue that specifying *normative requirements* (NR) is essential for these systems to be considered trustworthy, as it will prevent them from causing social, legal, ethical, empathetic, and cultural harms. However, eliciting such requirements is challenging, as it typically depends on a clear understanding of the system’s capabilities. This challenge is amplified in the case of large language model (LLM)-based chatbots, which operate as open-domain agents with an output space as vast as the natural language itself. As a result, identifying appropriate NR is difficult but critical to ensure these systems behave responsibly.

In this paper, we address this need by focusing on specific application domains and proposing contextualized NR prescribing how a chatbot should respond depending on the content of user inputs, context, or cultural expectations. As a starting point, we consider the Canadian context and examine three application domains that vary in their alignment with global guidelines, maturity within the AI and trustworthy AI literature, and real-world deployment levels: (i) therapy: a domain with well-established regional [1], [2] and global [3] ethical guidelines and significant attention from the AI and trustworthy AI communities (e.g., [4]–[9]). (ii) Education: a domain guided by global standards and of growing interest to AI and trustworthy AI researchers [10]–[16]. (iii) Customer service: a domain with extensive real-world deployment and company-specific or user-specific norms, yet limited formal research in the trustworthy AI literature despite the increasing prevalence of chatbot-based agents (e.g., agentic AI is projected to autonomously resolve up to 80% of routine customer service issues without human intervention by 2029 [17]). We chose to focus on Canada due to its diversity – comprising

ten provinces and three territories – and its complex regulatory landscape, which enabled an analysis of context-specific norms that vary across regional governments while remaining unified under some federal frameworks. We hope that by presenting NR across a diverse set of application domains, this exploratory study provides insights into how such requirements might be identified and mined.

The rest of this paper is organized as follows. We present the requirements by domain: healthcare in Sec.II, education in Sec.III, and customer service in Sec.IV. Finally, Sec.V summarizes our contributions and discusses remaining challenges.

II. THERAPY

A. Domain-Specific Norm Identification

To identify normative requirements for the domain of therapy, we first examined normative professional documents – such as codes of conduct, professional standards of practice, and codes of ethics – published by professional associations and regulatory bodies in Canada. In this country, the practice of psychology is regulated by individual provinces and territories, largely through regulatory colleges and boards [43]. Regulatory codes issued by these bodies are designed to ensure the protection of the public by providing clients with a clear, legally enforceable understanding of the standards governing licensed professionals’ conduct [21]. In contrast, documents developed and issued by professional associations often articulate the values, principles, and expectations that guide members in managing professional-client relationships [21], [44]. Their aim is to safeguard public welfare while promoting the profession’s integrity and ethical awareness. Although they may be informed by the structure and concepts of regulatory standards, they often adopt a broader scope, addressing both aspirational and advisory concerns as well as professional and regulatory matters [21], [30], [44]. Regulatory documents issued by government bodies may likewise refer to codes of ethics published by professional associations – either to explicitly adopt their content, to help clarify ambiguous provisions, or to provide supplementary ethical guidance [18], [21]. For all those reasons, we believe that such documents are relevant and suitable for defining therapy-related NR.

To identify documents of interest, we searched for “psychology professional standards of practice” and “psychology codes of ethics” for each province and territory in Canada.

TABLE I
NORMATIVE REQUIREMENTS FOR CHATBOTS ACROSS THREE DOMAINS*: THERAPY, EDUCATION, AND CUSTOMER SERVICE.

domain	normative requirements	references
Therapy	norm [†] <i>Accurately represent their own and their colleagues' qualifications in all spoken, written, or printed communications, being careful not to use descriptions or information that could be misinterpreted.</i>	[2], [18]–[29]
	rule T1 <i>When agent provides assistance, then specify that it is an LLM and to consult a medical professional.</i>	
	norm [†] <i>Present instructional information accurately, avoiding undue bias in the selection and presentation of information, and publicly acknowledge any personal values or bias that influence the selection and presentation of information.</i>	[2], [19]–[23], [28], [29]
	rule T2 <i>When agent provides assistance, then avoid undue bias in the selection and presentation of information.</i>	
	norm [†] <i>Counsellors/therapists limit their counselling/therapy services and practices to those which are within their professional competence by virtue of their education and professional experience, and consistent with any requirements for provincial/territorial and national credentials. They seek supervision, consult with and/or refer to other professionals when the counselling needs of clients exceed their level of competence.</i>	[18]–[22], [24]–[30]
	rule T3 <i>When user request is beyond the agent's competence, the agent must refer the user to competent professionals.</i>	
Education	norm [†] <i>Counsellors/therapists participate in only those practices that are respectful of the legal, civic, moral, and human rights of themselves and others, and act to safeguard the dignity and rights of their clients, students, supervisees, and research participants.</i>	[18], [20], [21], [30]
	rule T4 <i>When user request is not legal, civic, moral, or compliant with human rights, then agent must refrain from commenting.</i>	[23], [25]–[27], [29]
	norm [†] <i>Provides students with frequent and constructive feedback that allows them to track their learning and supports their progress.</i>	[31], [32]
	rule E1 <i>When providing feedback then the feedback must be constructive.</i>	
	norm [†] <i>To meet the learning needs of all students, teachers must take into account student variables such as demographics, maturity, and socioeconomic status.</i>	[33]
	rule E2 <i>When illustrating a concept then tailor it to align with individual student interests OR generation OR cultural background.</i>	
Customer Services	norm [†] <i>Teachers must have knowledge of and be able to design their instructional practices around students' interests, employing methodologies such as project-based or inquiry learning.</i>	[34], [35]
	rule E3 <i>When a participant expresses a challenge or issue, then help identify solutions collaboratively⁺.</i>	[33]–[37]
	rule E4 <i>When a participant expresses a challenge or issue, then do not provide direct solution.</i>	
	norm ^{†,◦} <i>Use interpretation service to take calls and in-person visits for customers who do not speak or have difficulty speaking and understanding English.</i>	[38]
	rule C1 <i>When user initiates conversation, the agent must ask for the user's preferred language of communication and respond using that language.</i>	
	norm ^{†,◦} <i>Staff are responsible for directing service requests and unresolved informal complaints to the appropriate department.</i>	[38]
	rule C2 <i>When user request is beyond the agent's competence, the agent must redirect the user request to an agent.</i>	
	norm [†] <i>The process for handling complaints will be transparent from the submission to the resolution stage, and complainants will be kept informed of the progress of their complaints and the outcomes achieved.</i>	[39], [40]
	rule C3 <i>When user submits a complaint, the agent must explain the complaint process and outline the expected timeline for resolution.</i>	
	norm ^{†,◦} <i>Our staff is dedicated to treating all customers with empathy, courtesy, and respect, acknowledging the value of every individual's perspective.</i>	[41]
	rule C4 <i>When a user expresses emotions such as frustration, disappointment, or anger, the agent must acknowledge the user's feelings and offer an apology.</i>	

*Each requirement is preceded by a direct or adapted fragment (labeled as *norm* and *in italics*) of the corresponding social or ethical norm; please find all fragments in [42].

[†]Ethical requirements derived from domain-specific codes of conduct and standards of practice (e.g., [2], [33]).

⁺In this work, identifying “solutions collaboratively” is interpreted as providing helpful hints, prompts, or engaging in brainstorming.

[◦]Empathetic requirements derived from domain-specific standards of practice, which emphasize the role of empathic engagement in justifying their normative guidelines (e.g., [38]).

With the exception of Nunavut, the search yielded 15 relevant documents: 12 regulatory documents corresponding to each applicable province and territory, a supplementary interpretive guide issued alongside the regulation in a specific province, and two codes of ethics issued by professional associations. Although many professional associations have generated codes of ethics, we primarily relied on two documents: the Canadian Code of Ethics for Psychologists, published by the Canadian Psychological Association (CPA)¹, and the Code of Ethics issued by the Canadian Counselling and Psychotherapy Association². Both codes outline ethical principles, values, and standards intended to guide members' conduct across all roles and communication modalities within the discipline. Many regulatory bodies and professional associations have formally adopted the CPA Code and reference it within their own codes of conduct, such that it constitutes a foundational component of the ethical framework governing Canadian psychologists.

B. From Domain-Specific Norms to Normative Requirements

To transform the domain-specific norms and standards into normative requirements, we reviewed the documents to

identify normative statements that could be adapted for use with LLMs. The original requirements extracted from these documents are presented in Column 3 of Table I. While a more extensive set of NR could have been considered, we chose to focus on a selected subset for two primary reasons. First, we focused on textual and conversational norms that could be applicable to LLM behavior, rather than those specific to humans and not readily transferable to LLMs (e.g., “Seek to design research, teaching, supervision, practice, and business activities in such a way that they contribute to the fair distribution of benefits to individuals and groups...” [2]). Second, we selected NR based on preliminary evaluations indicating that many LLMs failed to adhere to them. For these evaluations, which were undertaken for each domain of interest, we transformed posts and comments mined from social media platforms and online forums (e.g., Reddit) into user queries. These user queries then served as test prompts to assess the ability of six LLMs³ to comply with the requirements of interest. The NR presented in Table I reflect those that were consistently violated across models.

The normative requirements follow the “*when {trigger} then {response}*” format proposed by [51]. To convert the normative requirements into this format, we simplified and

¹The CPA is the largest national organization supporting psychological science, practice, and education in Canada

²The Canadian Counselling and Psychotherapy Association is a prominent bilingual association that advances the counselling and psychotherapy profession.

³The following LLMs were evaluated: GPT-4 [45], Claude 3 [46], Mistral [47], Gemini [48], DeepSeek [49], and LLaMA 3 [50].

reinterpreted the content to suit the operational context of LLMs. Specifically, we adjusted the requirements by removing elements related to the human, social, and operational dimensions of counselling – such as those involving personal biases, beliefs, or collaboration with colleagues – and to reflect the limitations and characteristics of LLMs (e.g., T2). In the process, many of the requirements were simplified and re-stated to reflect their underlying intention. Other ethical requirements related to therapeutic knowledge and competence were retained in the NR due to their applicability such that the final requirement is semantically similar to the original standards.

C. Real-World Therapy Chatbots

The use of generative AI in therapy – whether through platforms intentionally designed for mental health or those that serve this function incidentally – and its associated risks, underscore the promising role of NR in ensuring user safety. Character.ai [52], a popular platform that allows users to create chatbots based on real or fictional personas, offers evidence of this claim. On this platform, a bot titled *Psychologist* has amassed over 200 million interactions [53]. Although the platform is not dedicated to mental health, *Psychologist* is one of more than 400 bots related to therapy and psychology hosted on the site. While this, and similar tools [54], may offer mental health benefits to some users [55], the absence of appropriate guardrails introduces the risk of significant harm, including reported instances of suicide [56], [57]. In this context, the proposed NR (e.g., T1 and T3) offer a potential strategy to mitigate many of these negative consequences and ensure these systems align with ethical standards of practice, and are thus safe for users.

Despite the fact that many commercial and academic platforms have been developed to support mental health – some explicitly grounded in therapeutic principles – their relationship to the broader ethical standards of psychological practice remains unclear. For example, Woebot [58], Wysa [59], and Youper [60] employ cognitive behavioural therapy (CBT) techniques to help users manage their mental health, with the former two demonstrating effectiveness in reducing depressive symptoms [61], [62]. Complementing these commercial efforts, the research community has also proposed both general- and specific-purpose mental health chatbots. These include *Therabot* [63], which is grounded in CBT principles, as well as chatbots designed to foster self-compassion (e.g., *Vincent* [64]), encourage self-disclosure [65], [66], develop interpersonal skills (e.g., *Sunny* [67]), and alleviate stress (e.g., *Mylo* [68]). While aligning therapy chatbots with established therapeutic frameworks can help promote safety, the increasing sophistication and widespread use of LLMs also highlights the need to hold them to standards of practice similar to those governing human practitioners. The proposed NR may offer a practical path toward achieving this.

III. EDUCATION

A. Domain-Specific Norm Identification

To identify normative requirements for the domain of education, we reviewed professional standards of practice and codes of ethics issued by government bodies, regulatory agencies, and professional associations across Canada. In the Canadian context, education falls under provincial and territorial jurisdiction, with regional ministries, departments of education, and regulatory bodies responsible for key areas such as teacher certification and discipline [69]. As part of these responsibilities, many of these organizations publish standards and codes that articulate the professional ideals educators are expected to uphold – and to which they can reasonably, and often legally, be held accountable [34], [70], [71]. This regulatory framework is frequently supported by complementary documents developed by professional associations [72], [73]. Although distinct, association-issued standards and ethical guidelines often inform regulatory policies. In some cases, professional associations also serve as the first point of contact for addressing potential violations and refer matters to regulatory bodies when formal investigation is warranted [72], [73]. The use of these documents to identify and generate our NR thus ensures alignment with institutionally recognized norms, while also capturing the diversity inherent in the Canadian context.

To locate and collect the documents of interest, the authors searched for “teaching professional standards” for each Canadian province and territory. While the initial focus was on documents issued by government agencies and regulatory bodies, the search also returned materials published by national associations, which ultimately served to complement the regulatory sources. In total, 17 documents were identified and deemed relevant from this process based on the authority and legitimacy of the issuing body and the presence of rules that could be translated into NR for LLMs. With the exception of Prince Edward Island, documents were located for every Canadian province and territory, with some provinces yielding multiple sources. These materials included professional standards, codes of ethics and conduct, and reference frameworks outlining competencies or leadership practices. The education levels addressed in the 17 documents varied, spanning early childhood education to university instruction, such that they captured a broad spectrum of teaching practices and contexts.

B. From Domain-Specific Norms to Normative Requirements

To generate a preliminary list of NR for the education domain, we reviewed each document and extracted normative statements related to effective and ethical pedagogical practices. These statements, displayed in Column 3 of Table I, were selected for their specificity, interpretability, and the fact that during the preliminary evaluations discussed in Section II, many LLMs struggled to meet them. In translating these statements into NR, we employed a modular approach: complex practices were broken down into discrete components to improve clarity and enforceability. Additionally, some norms were refined to enhance interpretability for LLMs. For instance, the concept of “inquiry learning” was rephrased as

“identify solutions collaboratively”⁴ to provide clearer, action-guiding language.

C. Real-World Education Chatbots

Recent advancements in LLMs – and the promise of a revolution in education [74] – have been accompanied by a surge in AI-based tutors, highlighting the need to ensure these tools align with NR. Such tutors have been adopted by both established education technology companies and emerging platforms designed specifically for this purpose. For instance, Khan Academy’s AI chatbot, Khanmigo [75], guides learners through problem-solving rather than providing direct answers. Similarly, Duolingo has introduced adaptive lessons, AI roleplay, grammar explanations, and conversational practice with its AI agent Lily [76]. Alongside these enhancements to existing platforms, a wave of new applications has emerged to leverage AI’s diverse capabilities. Many offer step-by-step solutions and feedback on users’ work [77]–[79]. Others generate study materials directly from course documents, including flashcards, study guides, practice questions, and virtual tutoring [80]–[83]. While some services were developed with pedagogical principles in mind [75], many platforms have bypassed these considerations entirely. This has raised the specter of academic dishonesty [84] and concerns about overreliance on technology [85]. In this context, our proposed rules (e.g., E3 and E4) aim to help ensure that such tools fulfill the promises of constructive and pedagogically sound education chatbots.

IV. CUSTOMER SERVICES

A. Domain-Specific Norm Identification

In Canada, there is no single, overarching provincial or federal law that governs customer service standards. Instead, customer service is regulated through a combination of federal requirements, provincial legislation, industry-specific regulations, and voluntary codes of conduct. At the federal level, particular industries – such as telecommunications [86], banking [87], and air travel [88] – are subject to dedicated customer service standards and oversight. In contrast, most consumer interactions, including retail and personal services, are governed by provincial consumer protection laws which vary by jurisdiction (e.g., [89]–[91]). Many municipalities, the local level of government in Canada, have also created customer service standards to ensure clear, consistent practices that govern the interactions of municipal employees and stakeholders (e.g., customers, contractors) [38]–[41], [92]. In sectors that lack formal regulation, many organizations have also adopted voluntary standards and codes – such as the Scanner Price Accuracy Code [93] in retail and the Better Business Bureau’s Standards for Trust [94] – which promote fair and transparent customer service practices.

Given the diversity and scope of customer service practices in Canada, we focused on a narrow subdomain to make

⁴For the purpose of this work, “collaboration” is understood as providing hints, helpful prompts, or brainstorming.

the identification of NR more tractable: standards issued by municipal governments. We chose this subdomain for two primary reasons. First, the availability of multiple standards of practice across regions for a common use-case enabled the identification of generalizable NR. Second, the specified standards, in lieu of being strictly aspirational, provided clear, actionable guidance and were thus well-suited for translation into the relevant NR format [51]. Considering the large number of municipalities in Canada (5,162), we further limited their search to municipalities within a single province. To locate and collect the documents of interests, we searched for “customer service standards” in combination with the province’s name, which yielded a total of eight standards of practice. **Remark:** Identifying norms for the customer service domain was, in many respects, more challenging than for therapy and education. For the latter domains, we followed a relatively straightforward strategy: identify standards of practice and codes of ethics issued by government agencies, regulatory bodies, and national associations for each Canadian province and territory. In contrast, the customer service domain presented greater difficulty due to the fragmented nature of consumer protection regulations in Canada, which vary across provinces and territories, differ between federal and provincial jurisdictions, and often depend on the specific industry or type of good. To navigate this complexity, we focused on mining norms related to customer interactions, as these exhibited a degree of standardization and institutional support and aligned well with our overall methodological approach, leaving the extended set of norms as future work.

B. From Domain-Specific Norms to Normative Requirements

To generate the requirements from these standards, we reviewed each document to identify norms that were both commonly represented across sources and could be applicable to LLMs (Column 3 of Table I). Although many of these rules were specific to local governments, the authors selected norms that could generalize across sectors, and that would thus be applicable to a wide range of LLM-driven customer service contexts. To convert these NR into the relevant format [51], we applied three primary strategies. First, we abstracted away details tied to specific municipalities or organizational settings, focusing instead on the underlying goals and intentions of each requirement (e.g., C1). Second, we rephrased the norms to remove references to the human or social dynamics of customer service, thereby increasing their applicability to LLMs. Third, we introduced clarifying language where necessary to enhance specificity and improve interpretability for machine reasoning (e.g., C2).

C. Real-World Customer Service Chatbots

Modern LLM-powered service agents increasingly behave like trained human representatives rather than scripted FAQ bots. They sustain open-ended dialogue, remember context, and take action inside the same exchange. Ada’s virtual agent [95] asks clarifying questions, rephrases misunderstood queries, and reflects the brand policy in every interaction.

Amazon's Rufus [96] – a shopping assistant in the Amazon marketplace – helps customers with product search and recommendation. Gladly's Customer AI [97] implements customer support using AI chatbots, notifying the users about product/shipping updates and helps with returns/troubleshooting. Salesforce's Einstein Service Agent [98] goes further, interpreting intent, pulling CRM fields, and offering action buttons (e.g., "issue refund") so customers can resolve problems inside the chat itself. Start-ups push autonomy even further: Sierra AI [99] enables bots to execute multi-step workflows – verifying identity, updating subscriptions and arranging returns – while Palona AI [100] fully automates the interaction with the customer by conducting the dialogue, suggesting personal recommendations and placing the final order.

The growing deployment of AI chatbots in customer service has been accompanied by notable failures, underscoring the need for our proposed rules to help ensure trust and accuracy in these systems. For example, a customer service chatbot for a Canadian airline was found to hallucinate the company's bereavement policy [101], contrary to C1, while another airline company's chatbot inexplicably shared resources for a suicide hotline [102]. In this context, our proposed rules offer one mechanism to safeguard these AI systems from such errors. Indeed, many of our rules go further (e.g., C1 and C4 in Table I), helping ensure that these tools adapt to the user's language and treat all customers with empathy and courtesy. Rules such as C2 and C4 also support the responsible handling of customer complaints and unsupported requests.

V. CONCLUSION AND DISCUSSION

In this paper, we proposed twelve normative requirements that define how chatbot responses should behave to be ethical and empathetic across three application domains: therapy, education, and customer service. We detailed the process used to elicit these requirements for the Canadian context and discussed their application through examples of existing commercial chatbots expected to comply with them. We chose to focus on Canada, in part, because its federal system – characterized by distinct provincial and territorial jurisdictions – and its complex regulatory landscape enabled the analysis of context-specific norms that vary across regions. Indeed, Canada's system of government and its diversity – including ethnic, linguistic, and cultural dimensions – reflect those of many other federal systems and countries with hybrid legal and cultural frameworks, making it a particularly apt case study. In addition to considering other regions, our future work aims to develop a systematic methodology for eliciting normative requirements in open-domain systems, such as LLM-based chatbots, grounded in ethical and social norms. Going forward, we also aim to investigate which requirements can be directly reused, shared, or adapted across domains, which are domain-specific, and how high-level NR can be operationalized into concrete, testable forms that enable automated assessment of LLM compliance. Finally, we will explore whether deployed chatbots adhere to these requirements and what it would take to enforce compliance when they do not.

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